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PALEOMAGNETISM OF PALEOZOIC ASPHALTIC DEPOSITS IN SOUTHERN OKLAHOMA, USA

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Abstract. Paleomagnetic measurements on asphaltic samples from two formations in southern Oklahoma have been performed. A bioclastic unit from the Boggy Formation, known as the Buckhorn asphalt, exhibited a stable, characteristic remanent moment (RM) after A.F. demagnetization between 5-20 mT. We infer from our data that very fine, possibly authigenic magnetite, like that shown to have a genetic relationship with the migration through rocks of hydrocarbons [Elmore et al.,1987], is the primary RM carrier in these samples. The tilt corrected paleopole for the Buckhorn asphalt (121.9 E; 43.5 N; δ_p= 1.3; δ_m= 2.3) falls on the Early Permian Apparent Polar Wander Path for North America of Irving and Irving [1982], using a 30 Ma window (270-280 Ma). Because the Boggy Form-ation, containing the Buckhorn asphalt, was depos-ited during the Late Pennsylvanian (Desmoinesian), we interpret the data to indicate magnetization during minor uplift in the Early Permian of the Arbuckle Mountain region. The RM appears to have been acquired at this time, probably as the result of magnetite production facilitated by the introduction time of sulfate reducing bacteria.

Introduction

Previous paleomagnetic measurements on rocks containing hydrocarbons, such as the Athabasca Oil Sands in Alberta, Canada [Ellwood and Pemberton, 1984] and some Utah oil shales [Richardson and Noltimier, 1980], have demonstrated that these rocks may carry a stable remanent moment (RM). Recent work by Elmore et al., [1987] has shown that very fine grained (<3 μm), and abundant authigenic magnetite was produced during hydrocarbon migration through Permian limestones in southern Oklahoma. These magnetite grains, associated with hydrocarbons are expected to carry a stable RM.

At The University of Texas at Arlington (UTA), we have been examining the magnetic properties of rocks from the Arbuckle Mountain region, for the purposes of joint magnetostratigraphic-biostratigraphic studies. Our results have been disappointing, primarily because these rocks appear to have been remagnetized. This remagnetization may be due to a number of factors, including the oxidation of unstable minerals, iron oxides, sulfides and carbonates, leading to the development of stable but secondary RM carriers. Important among these minerals is the iron carbonate, siderite, which has been shown to cause substantial RM changes, including self-reversals in samples during measurement in the laboratory [Ellwood et al., 1986]. It has also been inferred that, during dedolomitization, other iron carbonates such as ferroan dolomite can produce secondary iron oxides which may dominate the RM [Elmore et al., 1985].

In an attempt to locate samples which were protected from oxidation and therefore remagnetization, we decided to sample asphaltic deposits from two localities near the city of Sulfur in the Arbuckle Mountains of southern Oklahoma. The asphalts, identified as Middle Ordovician and Upper Pennsylvanian in age, are exposed and have been quarried [Ham, 1973]. One of these, the Buckhorn asphalt (FL-2), contains a suite of very well preserved fossils (including at least one genus of a heretofore undescribed insect), and primary aragonite, normally unstable, is preserved in the asphalt [Crick and Ottensman, 1983]. Crick and Ottensman [1983] argue that the presence of the aragonite indicates that the hydrocarbons in the deposit must have intruded soon after deposition of the sediments.

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Geological Setting, Sampling and Methods

A series of small diameter cores were recovered using a portable gasoline powered drill, from each asphalt quarry. In the first (FL-1), the asphalt is contained in the basal sand unit of the Oil Creek Formation (Chazyan - Middle Ordovician). The quarry is located ~5 km southwest of Sulfur, Oklahoma, along the eastern edge of the Arbuckle Mountain anticline. The sand is extremely well sorted and uncemented except for the binding characteristics of the asphalt. The second quarry (FL-2) is located in the Boggy Formation, and the deposit is commonly referred to as the Buckhorn asphalt (Desmoinesian - Pennsylvanian). It lies ~1 km south of locality FL-1. The asphaltic portion of the Boggy Formation consists of a matrix of asphalt which serves to lithify a coquina of molluscan shell debris deposited in a series of 5-10 cm thick beds. Bedding at quarry FL-1 was measured to dip 10° along an azimuth of 330°. At site FL-2, mean bedding was measured to dip 27° along an azimuth of ~290°. Cores were oriented using a Brunton compass, marked and returned to the laboratory where ~2.2 cm long specimens were cut from each core.

The RM was measured after A.F. demagnetization to 100 mT magnetic induction for pilot samples, using a Molspin Minispin spinner magnetometer. Vector demagnetization plots [Zijderveld, 1967] were constructed from these data (Figure 1) and evaluated. We decided, from this examination, that the natural RM (NRM) for all samples would be A.F. demagnetized to 30 mT induction levels.

An ARM was imparted to several samples from

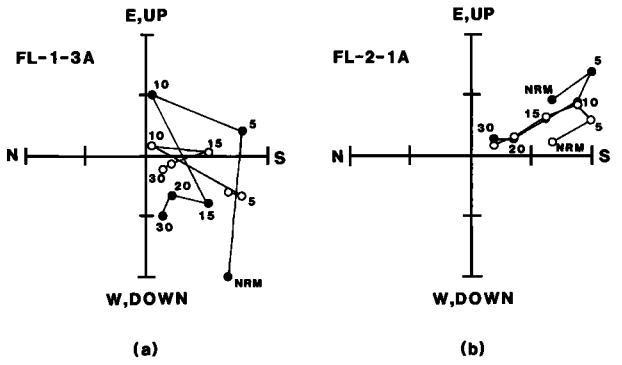


Fig. 1. Normalized A.F. demagnetization plots (Zijderveld, 1967) of RM directions from typical asphalt samples at quarry FL-1 (Figure 1a) and FL-2, the Buckhorn Asphalt quarry (Figure 1b). Filled circles represent projections on a horizontal plane. Open circles are plotted on a vertical plane. Total moment ranges from 1.4 x 10^{-3} to 9.3 x 10^{-5} Am^2. Demagnetization levels in millitesla (mT). Note that the FL-1 samples show no consistent demagnetization trends, while Buckhorn Asphalt samples show consistent RM demagnetization decay paths.

both quarries following A.F. demagnetization. Thermal and further A.F. demagnetization experiments were performed on these samples using a Schonstedt TSD-1 Thermal Specimen Demagnetizer and A.F. 2 axis tumbler, respectively. As expected, heated samples began to melt, boil and seriously deform at low temperatures (100-200°C). While this deformation produced large changes in ARM direction, total moment remained strong (1 x 10^{-3} Am^2) after heating to 400°C (Figure 2), even though some of the asphalt flowed out of the samples and pooled in the thermal boat.

Anisotropy of magnetic susceptibility (AMS) measurements in low fields (<10 mT) were conducted on selected undeformed asphaltic samples using torsion fiber magnetometers (see King and Rees [1962] and Nye [1969] for a discussion of the method). Thin sections were prepared from hand samples taken from the Buckhorn asphalt. X-ray diffraction (XRD) analyses were performed using a Philips X-ray diffractometer utilizing CuK α radiation.

Results

Optical examination of these samples did not indicate the presence of iron oxides, but XRD analysis did show the major spectral lines for pyrrhotite. We could not differentiate between magnetic versus the non-magnetic pyrrhotite. Since pyrrhotite is known to occur as an authigenic mineral in reducing sedimentary environments [Berner, 1964] and may exhibit ferrimagnetic behavior, the magnetic data reported here might be interpreted to indicate that pyrrhotite is a contributor to the primary carrier of the RM in the rock. However, the thermal demagnetization experiments show that an RM is relatively stable after thermal demagnetization (Figure 2) to temperatures above the Curie points for most magnetic phases of pyrrhotite (~320°C) or other magnetic iron sulfides (e.g. greigite ~350°C). Furthermore, the median destructive temperature in these samples appears to lie above 300°C. Therefore, we suggest that the probable carrier of most of the characteristic RM in these samples is magnetite, even though pyrrhotite may be present in large enough concentrations to identify, using XRD analysis.

Samples from quarry FL-1, composed of loosely consolidated sand with an asphaltic matrix, exhibited RM directions for which no stable or characteristic RM could be identified (Figure 1a). However, during demagnetization and RM measurement, the FL-1 asphalt was observed to flow slightly in the sand matrix (this was not observed for the bioclastic samples from the Buckhorn asphalt), and is the probable reason for the RM scatter. It is interesting to note that magnetite spheres like those reported by Elmore et al.[1987] have been isolated from the Oil Creek quarry (Elmore, personal comm.) and are probably the RM carrier in both FL-1 and FL-2 samples. A.F. demagnetization data of an ARM for an FL-1 sample show median destructive induction levels higher than for FL-2 samples (Figure 2).

Samples from the Buckhorn Asphalt quarry (FL-2) exhibit good between-sample, within-site consistency (Figure 3) after A.F. cleaning to 5-10 mT induction, and the corresponding removal of a secondary RM overprint (Figure 1b). Linear decay

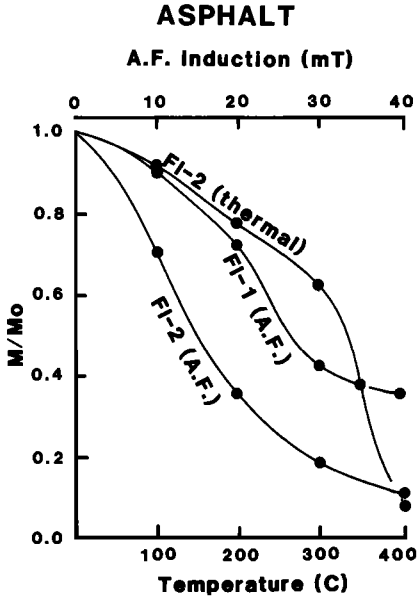


Fig. 2. Normalized thermal and A.F. demagnetization of an ARM imparted to asphaltic samples (FL-2), and normalized A.F. demagnetization for an ARM in an FL-1 sample. Temperature to 400°C and magnetic induction to 40 mT. Samples were given an ARM in a 1 mT D.C. field before demagnetization. We attribute the sharp decrease in RM, following thermal demagnetization above 300°C, to flow of the asphalt out of the sample thus disrupting the RM. Total moment after 400°C was still high, 1 x 10^{-3} Am^2.

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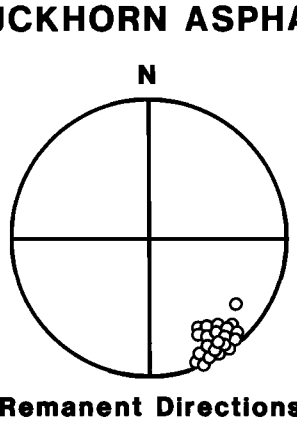


Fig. 3. Equal area plot of RM directions after 10 mT A.F. cleaning of the NRM for all Buckhorn Asphalt samples (FL-2; n= 55). Directions have not been tilt corrected. Open circles represent negative inclinations.

trends toward the origin are observed with magnetic cleaning to 10-25 mT (Figure 1b). At greater induction levels (>25 mT), the RM directions begin to diverge from consistent trends. This scatter is probably due to the very low total moments remaining in samples above 25 mT (generally <2 x 10^{-4} Am^2).

Discussion

Because of the very limited exposure of these sediments, ~300 m^2, a collection representing a large area was not possible. However, the cores collected from the Buckhorn asphalt, without the rejection of data from any samples, yield a uniform RM mean direction with excellent precision (D = 146.1, I = -11, α_{95} = 2.1). The uncorrected paleopole calculated for this direction exhibits slightly lower latitudes than do poles for North America during the Late Paleozoic [Irving and Irving, 1982]. After a tilt correction is applied to the data, the paleopole (43.5° N, 121.9° E, δ_p = 1.3, δ_m = 2.3) falls on the apparent North American polar wander path (APWP) of Irving and Irving [1982] between 270-280 Ma (Early Permian), using their 30 Ma smoothing window. The Buckhorn pole is also essentially identical with the pole reported by Helsley [1965] for the early Permian Dunkard Series rocks of West Virginia. One might argue that calibration of the Mid-Late Paleozoic APWP may not be accurate enough to differentiate, with certainty, an RM acquired in the Late Pennsylvanian from that acquired in the Early Permian. Nevertheless, our data from the Late Pennsylvanian Boggy Formation exhibit a tilt corrected pole which appears to have been acquired in the Early Permian.

The evidence leads us to believe that acquisition of an RM in the Buckhorn Asphalt probably resulted from uplift following deposition. The paleopole is consistent with an Early Permian magnetization acquired during minor, post-deposition uplift in southern Oklahoma [Ham, 1973]. This event appears to be relatively minor because the matrix containing the asphalt exhibits only slight changes resulting from deformation, such as local, but limited calcite recrystallization of the fragile interiors of some mollusk tests. The AMS data show trends (Figure 4) which are consistent with the near horizontal principal stresses, oriented along a NNW-SSE azimuth, which are known to have affected the southern Oklahoma area in the Early Permian [Ham, 1973]. It is possible that uplift at this time mobilized the magnetite in these samples, thus allowing a new RM to be acquired. However, we believe that a better explanation for the data is that magnetite production was facilitated by the introduction of sulfate reducing bacteria at the time of uplift and as a result of near surface exposure of the deposit.

It is our interpretation of the data, that the migration of hydrocarbons into the Boggy Formation must have occurred in the Late Pennsylvanian to preserve the aragonite in the deposit. Later, due to the high volatile content and low viscosity of the hydrocarbons at a time of minor uplift, probably in the Early Permian, internal mixing with ground water and associated fluids may have been relatively active. Assuming that these deposits were nearly exposed at this time, then sulfate reduction, probably controlled by bacteria [Berner, 1981] may have produced pyrrhotite (observed above hydrocarbon reservoirs such as the Cement Field in Oklahoma [Reynolds et al., 1985]) in a similar manner to pyrrhotite

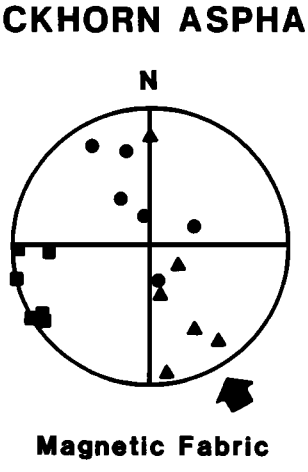


Fig. 4. Equal area lower hemisphere plot of samples from the Buckhorn Asphalt quarry (FL-2). Arrow represents the direction of principal horizontal stress estimated for the study area (Ham, 1973). Squares = maximum (χ_1), triangles = intermediate (χ_2), and circles = minimum (χ_3) susceptibility axes. Estimation of a stress direction is based on the work by Graham (1966), who showed that with progressive (but not excessive) deformation, a sediment will deform so that the χ_1 dimension of the AMS ellipsoid will rotate toward a direction normal to the direction of principal stress. Furthermore, at low to intermediate deformation magnitudes, the χ_2 and χ_3 azimuths will describe a great circle parallel with the principal stress. These AMS distributions are apparent in the Buckhorn data.

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production in marine sediments [Berner, 1964]. It has been demonstrated that hydrocarbon migration through (or into) sediments may also result in the growth of very small magnetite grains (<3 μm [Elmore et al., 1987]), and the development of a CRM. Thermal and A.F. demagnetization trends (Figure 2a) indicate that the RM, which is probably a chemical RM (CRM), is carried by fine magnetite grains. Since the hydrocarbons arrived earlier, production of magnetite probably occurred in the Buckhorn asphalt before devolatilization. We suggest that later devolatilization significantly decreased the viscosity of the fluid, effectively halting circulation and water penetration into the now viscous asphalt, thus protecting the magnetite from weathering and oxidation and preserving the CRM. Later, small amplitude tilting and deformation produced the bedding orientations now observed.

Conclusion

Although the data are limited, it is clear that some asphaltic deposits do yield a relatively stable characteristic magnetization which can be used to assess the time of development or time of deformation of the deposit. While the coarse grained sediments containing asphalt at site FL-1 did not yield interpretable results, the bioclastic unit known as the Buckhorn asphalt exhibited a characteristic RM. We interpret the data to indicate that a CRM was acquired by this deposit, probably in the Early Permian, during or immediately following minor uplift of the Upper Pennsylvanian host rock. We suggest that uplift of the Arbuckle Mountain region, and associated erosion, brought the deposit close enough to the surface that ground waters were able to interact with the hydrocarbons, introducing the sulfate reducing bacteria responsible for growth of the magnetic carrier, inferred to be very fine (<3 μm) magnetite grains. Devolatilization of the hydrocarbons produced the asphalt we see today by increasing the viscosity, thus reducing ground water interaction and preventing magnetite oxidation.

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